

LESSON 7-5 • HOMEWORK PACKET • SHEET 1 OF 2

Plotting Points on the Coordinate Plane

Maryland Standard 6.NS.C.6c • Learning target – I can plot a point with rational coordinates and name the quadrant it lies in.

REMEMBER

- Start at the **origin** (0, 0). Move x units right or left, then y units up or down.
- Quadrant signs: I (+, +), II (–, +), III (–, –), IV (+, –).
- A point with a 0 coordinate is **on an axis**, not in a quadrant.

WORKED EXAMPLE

Plot $(-3.5, 2)$ and name its quadrant.

Step 1 Start at the origin. Move left 3.5 units.

Step 2 Move up 2 units. Left and up is Quadrant II.

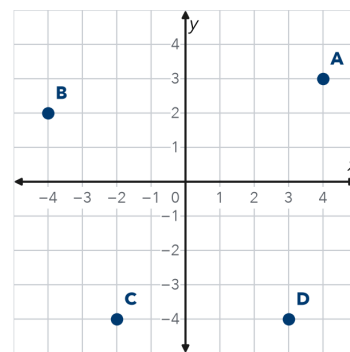
Answer Quadrant II

PRACTICE

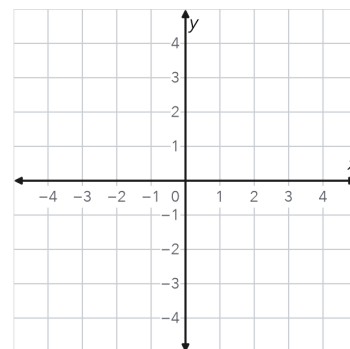
show your work • label units

1. Write the ordered pair for each point. Name its quadrant.

Point	Ordered pair	Quadrant
A		
B		
C		
D		



2. **Town Map.** The library is at the origin. Plot and label each place: Park P(2, 4), Pool Q(–4, –1), School S(–3.5, 3), Garden G(4, –2.5). Which place is in Quadrant III? _____ Quadrant IV? _____



3. Name the quadrant or axis for each point.

a. $(-7, -2)$ _____

b. $(8, -1)$ _____

c. $(0, 6)$ _____

d. $(-1.5, 4)$ _____

e. $(-9, 0)$ _____

f. $(2.5, -6)$ _____

4. **Reasoning.** Malia says $(-3, 5)$ and $(5, -3)$ name the same point. Is she right? Explain.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME With someone at home, name a quadrant. Take turns saying an ordered pair that lands in it.

LESSON 7-5 • HOMEWORK PACKET • SHEET 2 OF 2**Reflecting Points Across Axes**

Maryland Standard 6.NS.C.6b • Learning target – I can reflect a point across the x-axis or the y-axis and name the new coordinates.

REMEMBER

- **x-axis** reflection: flip the sign of y . $(3, 4)$ becomes $(3, -4)$.
- **y-axis** reflection: flip the sign of x . $(3, 4)$ becomes $(-3, 4)$.
- Across **both** axes, both signs flip.

WORKED EXAMPLE

Reflect $(-2, 5)$ across the x-axis.

Step 1 Keep $x = -2$.

Step 2 Change the sign of y : 5 becomes -5 .

Answer $(-2, -5)$

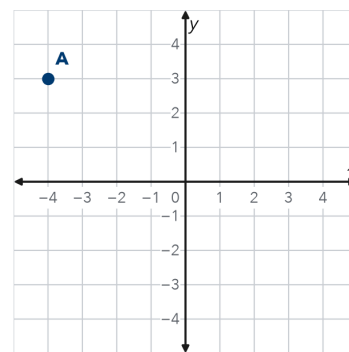
PRACTICE

show your work • label units

1. Point A is at $(-4, 3)$. Plot point B, the reflection of A across the x-axis. Plot point C, the reflection of A across the y-axis.

a. B = _____

b. C = _____



2. Write the new point after each reflection.

a. $(-6, -2)$ across the y-axis _____

b. $(-4, 7)$ across both axes _____

3. **Park Map.** A bench is the reflection of a bus stop at $(7, -2.5)$ across the x-axis. Where is the bench?

Answer _____

4. **Stretch.** Point K was reflected across the x-axis. The new point is $(-6, -5)$. Where was point K?

K _____

5. **Find the Error.** Sam reflected $(-3, -8)$ across the y-axis and got $(-3, 8)$. Explain his mistake. Fix it.

TEST PRACTICE

choose the best answer

6. Point M $(-5, 2)$ is reflected across the y-axis. What is the new point?

A $(-5, -2)$

B $(5, 2)$

C $(5, -2)$

D $(2, -5)$

7. The points $(6, -1)$ and $(6, 1)$ are reflections of each other across –

A the x-axis

B the y-axis

C both axes

D neither axis

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about # _____